



10159 - Using the Einstein zigzag lens to break the Mass-Sheet Degeneracy.

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) J1721+8842

ABSTRACT

We propose to exploit the unique lensing configuration of J1721+8842, the Einstein zigzag lens, to measure the Hubble constant using time-delay cosmography, a single-step method independent of any other cosmological probes. J1721+8842 is an exceptional six-image lensed quasar, lensed twice by two intervening galaxies at $z=1.8845$ and $z=0.1840$. The nearly perfect alignment of the two deflectors produces two additional images of

the background quasar at $z=2.38$, with light following a characteristic zigzag optical path between the galaxies.

This system will deliver the most precise measurement of H_0 from an individual lens (4.5% precision), sufficient to distinguish, at the 2 sigma level, between early (CMB-based) and late (distance-ladder) measurements. It will also provide new insights into the mass distributions of lens galaxies, thereby reducing one of the main sources of residual systematics in time-delay cosmography. The results will be directly applicable to larger lens samples.

Specifically, we will use JWST/NIRSpec IFU observations to measure the velocity dispersion of the second deflector and construct resolved kinematic maps of the lensing galaxy at $z=1.8845$. These data will break the multi-lens-plane generalization of the mass-sheet degeneracy (MSD), the dominant source of uncertainty in time-delay cosmography. As demonstrated in this proposal, the kinematics of the second deflector provide orthogonal constraints on the mass-sheet degeneracy parameter and will therefore substantially enhance the scientific impact of existing Cycle 2 JWST/NIRSpec observations, which targeted the first deflector of this system.

OBSERVING DESCRIPTION

We request deep spectroscopic observations with NIRSpec IFU of the second deflector in the "Einstein zigzag" lens, J1721+8842, at redshift 1.8845. This exceptional strong-lensing system forms six images of a quasar at redshift 2.38, which is lensed by two galaxies at $z=0.1840$ and $z=1.8845$. The second deflector, targeted in this proposal, is itself lensed into an extended arc by the first deflector.

These data will enable spatially resolved kinematics, allowing us to break the double-lens-plane generalization of the mass-sheet degeneracy (MSD) and measure the Hubble constant with unprecedented precision from a single system. We will use the G140M/F100LP disperser/filter setup to target rest-frame optical absorption features (Ca H & K and others). The pointing is centered on the second deflector, complementing existing data that targeted the first deflector. A standard four-point dither pattern will be applied to mitigate defects and cosmic rays, and the NRSIRS2 readout pattern will be used, as recommended for long integrations on faint objects. The faintest quasar image is also targeted to obtain a high-SNR template for subtraction from the lens galaxy spectra, reducing QSO contamination. The exposure time is calculated to achieve $\text{SNR}=20$ per $\text{\AA}$ per Voronoi bin, sufficient to measure stellar velocity dispersions with 5% precision and subpercent covariance between bins and systematic errors. These data will constrain the mass-sheet degeneracy parameter to 3.8% precision, which, combined with existing data and lensing constraints, will enable a precision of 4.5% on H_0 for this single system, including the MSD and all other sources of uncertainties.

Proposal 10159 - Targets - Using the Einstein zigzag lens to break the Mass-Sheet Degeneracy.

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	J1721+8842	RA: 17 21 41.9981 (260.4249921d) Dec: +88 42 22.65 (88.70629d) Equinox: J2000 <i>Comments: Coordinates correpond to the center the second deflector.</i> <i>Category=Galaxy</i> <i>Description=[Einstein rings, Elliptical galaxies]</i> <i>Extended=YES</i>		

Proposal 10159 - Observation 1 - Using the Einstein zigzag lens to break the Mass-Sheet Degeneracy.

Observation	Proposal 10159, Observation 1												Fri Mar 13 19:00:45 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Comments: PA requirement needed to minimize QSO contamination within the IFU FOV.												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	J1721+8842	RA: 17 21 41.9981 (260.4249921d) Dec: +88 42 22.65 (88.70629d) Equinox: J2000										
	Comments: Coordinates correpond to the center the second deflector.												
	Category=Galaxy												
	Description=[Einstein rings, Elliptical galaxies]												
Extended=YES													
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Ex p	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	25	3	false	true	NONE	4	12	22058.402	145108	
Special Requirements	Aperture PA Range 35.9 to 61.9 Degrees (V3 256.92835083 to 282.92835083) Aperture PA Range 125.9 to 151.9 Degrees (V3 346.92835083 to 12.92835083) Aperture PA Range 215.9 to 241.9 Degrees (V3 76.92835083 to 102.92835083) Aperture PA Range 305.9 to 331.9 Degrees (V3 166.92835083 to 192.92835083)												